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QUESTIONS



Deep Learning Fundamentals Lab Admissions Quiz

Question 1

What does the transpose operation do to a matrix?

- ☐ Sorts elements in ascending order
- ☐ Negates all elements
- ☐ Inverts all elements
- ☒ Swaps rows and columns

Question 2

Why can't you multiply a 2×3 matrix by a 2×4 matrix?

- ☐ First matrix must have more columns
- ☒ Inner dimensions don't match ($3 \neq 2$)
- ☐ The matrices must be square
- ☐ They have different total elements

Question 3

Calculate the trace of $\begin{bmatrix} 3 & 1 & 2 \\ 0 & 5 & 1 \\ 2 & 3 & 4 \end{bmatrix}$

- ☐ 21
- ☐ 9
- ☐ 7
- ☒ 12

Question 4

Which operation is valid for any two matrices A and B of the same size?

- ☐ Multiplication
- ☐ Division
- ☒ Addition
- ☐ Determinant of A+B

Question 5

Why can a system of linear equations have infinitely many solutions?

- ☒ When equations are dependent with free variables
- ☐ Only happens with infinite variables
- ☐ When there are infinite equations
- ☐ When all coefficients are infinite

Question 6

Evaluate: "Every matrix has at least one eigenvalue." Is this always true?

- ☒ False. Some matrices have no eigenvalues

- ☐ True for all matrices including rectangular
- ☐ True for square matrices over complex numbers
- ☐ False. Only symmetric matrices have eigenvalues

Question 7

What is the chain rule used for?

- ☒ Finding derivatives of composite functions
- ☐ Adding derivatives together
- ☐ Integrating products
- ☐ Finding second derivatives

Question 8

Why is $f(x) = |x|$ not differentiable at $x = 0$?

- ☐ The function is discontinuous at 0
- ☐ The limit doesn't exist at 0
- ☐ The function is undefined at 0
- ☒ Left and right derivatives are different

Question 9

Calculate $\frac{d}{dx}[e^{3x}]$

- ☐ $3xe^{3x-1}$
- ☐ e^{3x}
- ☒ $3e^{3x}$
- ☐ e^3

Question 10

Calculate the gradient of $f(x, y) = x^2 + 2xy + y^2$ at $(1, 1)$

- ☐ $(1, 1)$
- ☐ $(2, 2)$
- ☐ $(3, 3)$
- ☒ $(4, 4)$

Question 11

How do you determine concavity using derivatives?

- ☐ Find where function is positive
- ☒ Check the sign of the second derivative
- ☐ Check where first derivative is zero
- ☐ Calculate the third derivative

Question 12

Evaluate: "Every continuous function has an antiderivative." Is this correct?

- ☐ False. Some can't be integrated
- ☒ True by Fundamental Theorem of Calculus
- ☐ False. Need differentiability
- ☐ True only for polynomials

Question 13

What's the difference between population and sample?

- ☐ No difference
- ☐ Sample is always larger
- ☒ Population is all data sample is a subset
- ☐ Population is theoretical

Question 14

What's the difference between discrete and continuous distributions?

- ☐ Discrete is finite continuous is infinite
- ☒ Discrete has countable outcomes continuous has intervals
- ☐ Continuous is more accurate
- ☐ No real difference

Question 15

What type of distribution models coin flips?

- ☐ Uniform
- ☐ Exponential
- ☒ Binomial
- ☐ Normal

Question 16

If you flip a fair coin 3 times, what's $P(\text{exactly 2 heads})$?

- ☐ $1/2$
- ☐ $1/4$
- ☒ $3/8$
- ☐ $2/3$

Question 17

Why might correlation not imply causation?

- ☐ Causation doesn't exist
- ☐ Statistics are unreliable
- ☐ Correlation is always wrong
- ☒ Confounding variables may exist

Question 18

Evaluate: "Larger samples always give better estimates." True or false?

- ☐ Always true
- ☒ True if sampling is unbiased
- ☐ True only for normal data
- ☐ Always false

Question 19

What keyword starts a class definition in Python?

- ☒ `class`
- ☐ `type`
- ☐ `def`

☐ object

☐ struct

Question 20

How do you create an empty dictionary?

☐ []

☒ {} or dict()

☐ ()

☐ {None}

Question 21

What's the scope of a variable defined inside a function?

☐ Global everywhere

☐ Deleted after one use

☒ Local to that function

☐ Available in all functions

Question 22

How are == and is different?

☒ == checks if values are equal, is checks if they're the same object

☐ == only works on numbers

☐ is is faster

☐ They're identical

Question 23

Why might a while loop be preferred over a for loop?

☐ While loops use less memory

☐ While loops are always faster

☐ For loops can't use conditions

☒ When iteration count is unknown

Question 24

Create a class with a property that can't be modified from outside.

☐ Use 'const' declaration

☐ Use 'private' keyword

☐ Make it a class variable

☒ Use underscore prefix and @property decorator

SUBMIT

GO BACK

SAVE & CONTINUE

